

Д/з по теме "определители"

1.2.59

$$\begin{vmatrix} 2 & 3 \\ 5 & -4 \end{vmatrix} = 2 \cdot (-4) - (-5) \cdot 5 = -8 - (-15) = 7$$

1.2.60

$$\begin{vmatrix} \alpha & \beta \\ 0 & 0 \end{vmatrix} = \alpha \cdot 0 - \beta \cdot 0 = 0$$

1.2.61

$$\begin{vmatrix} x^2 & x \\ xy & y^2 \end{vmatrix} = x^2 \cdot y^2 - xy^2 \cdot x = x^2 y^2 - x^2 y^2 = 0$$

1.2.62

$$\begin{vmatrix} \alpha & 3\alpha \\ \beta & 3\beta \end{vmatrix} = 3\alpha\beta - 3\alpha\beta = 3\alpha\beta - 3\alpha\beta = 0$$

1.2.63

$$\begin{vmatrix} \cos \varphi & \sin \varphi \\ \sin \varphi & \cos \varphi \end{vmatrix} = \cos \varphi \cdot \cos \varphi - \sin \varphi \cdot \sin \varphi = \cos^2 \varphi - \sin^2 \varphi = \cos 2\varphi$$

1.2.64

$$\begin{vmatrix} x & x-1 \\ x^2+x+1 & x^2 \end{vmatrix} = x^2 - (x-1)(x^2+x+1) = x^2 - (x^3 - x^2 + x^2 - x + x - 1) = x^2 - x^3 + x^2 - x^2 + x - x + 1 = x^2 - x^3 + 1 = -1$$

1.2.65

$$\begin{vmatrix} 2x-5 & 4 \\ -x & -3 \end{vmatrix} = 0$$

$$\begin{aligned} -5(2x-5) - 4(-x) &= 0; \\ -6x + 25 + 4x &= 0; \\ -2x &= -9; \\ x &= 4,5 \end{aligned}$$

1.2.66

$$\begin{vmatrix} x-2 & x+1 \\ x-1 & x-2 \end{vmatrix} = 0$$

$$\begin{aligned} (x-2)(x-2) - (x+1)(x-1) &= 0 \\ (x-2)^2 - (x^2-1) &= 0 \\ x^2 - 4x + 4 - x^2 + 1 &= 0 \\ -4x &= -5 \\ x &= 1,25 \end{aligned}$$

1.2.67

$$\begin{vmatrix} 3-x & x+2 \\ x+1 & x-1 \end{vmatrix} = 6$$

$$\begin{aligned} (3-x)(x-1) - (x+2)(x+1) &= 6 \\ 3x - 3 - x^2 + x - (x^2 + x + 2x + 2) &= 6 \\ 3x - 3 - x^2 + x - x^2 - x - 2x - 2 &= 6 \\ -2x^2 + x - 5 - 6 &= 0 \\ -2x^2 + x - 11 &= 0 \quad | \cdot (-1) \\ 2x^2 - x + 11 &= 0 \\ D = 1 - 4 \cdot 2 \cdot 11 = 1 - 88 = -87 < 0 \Rightarrow \end{aligned}$$

$\Rightarrow x \notin \mathbb{R}$

1.2.68

$$\begin{vmatrix} x-2 & y+3 \\ 1-y & x-2 \end{vmatrix} = -4$$

$$\begin{aligned} (x-2)^2 - (1-y)(y+3) &= -4 \\ x^2 - 4x + 4 - (y + 3 - y^2 - 3y) + 4 &= 0 \\ x^2 - 4x + 4 - y - 3 + y^2 + 3y + 4 &= 0 \\ x^2 - 4x + 5 + y^2 + 2y &= 0 \\ x^2 - 4x + 4 + y^2 + 2y + 1 &= 0 \quad | +1 \\ (x-2)^2 + (y+1)^2 &= 0 \\ \begin{cases} (x-2)^2 = 0 \\ (y+1)^2 = 0 \end{cases} \Rightarrow \begin{cases} x = 2 \\ y = -1 \end{cases} \end{aligned}$$

1.2.69

$$\begin{vmatrix} x-2 & y+3 \\ 7-y & x+4 \end{vmatrix} = -24$$

$$\begin{aligned} \frac{(x-2)(x+4) - (y+3)(7-y)}{x^2 + 4x - 2x - 8 - 7y - y^2 + 21 - 3y} + 34 &= 0 \\ x^2 + 4x - 2x - 8 - 7y - y^2 + 21 - 3y + 34 &= 0 \\ x^2 + 2x + y^2 - 4y + 5 &= 0 \\ x^2 + 2x + 1 + y^2 - 4y + 4 &= 0 \\ (x+1)^2 + (y-2)^2 &= 0 \\ \begin{cases} (x+1)^2 = 0 \\ (y-2)^2 = 0 \end{cases} \Rightarrow \begin{cases} x = -1 \\ y = 2 \end{cases} \end{aligned}$$

1.2.70

$$\begin{vmatrix} \sin 2x & -\sin 3x \\ \cos 2x & \cos 3x \end{vmatrix} = 0$$

$$\begin{aligned} \sin 2x \cos 3x + \sin 3x \cos 2x &= 0 \\ \sin(2x+3x) &= 0 \\ \sin 5x &= 0 \end{aligned}$$

$$5x = \pi n, n \in \mathbb{Z} \quad | \cdot \frac{1}{5}$$

$$x = \frac{\pi n}{5}, n \in \mathbb{Z}$$

1.2.71

$$\begin{aligned} \rightarrow \begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 3 \\ 4 & 6 & 7 \end{vmatrix} &= 1 \cdot \begin{vmatrix} 3 & 3 \\ 6 & 7 \end{vmatrix} - 1 \cdot \begin{vmatrix} 2 & 3 \\ 4 & 7 \end{vmatrix} + 1 \cdot \begin{vmatrix} 2 & 3 \\ 4 & 6 \end{vmatrix} = \\ &= 21 - 18 - (14 - 12) + 12 - 12 = 3 - 2 = 1 \end{aligned}$$

+ - +
- + -
+ - +

1.2.72

$$\begin{aligned} \rightarrow \begin{vmatrix} 1 & 1 & 0 \\ 2 & 3 & 1 \\ 0 & 2 & 3 \end{vmatrix} &= 1 \cdot \begin{vmatrix} 3 & 1 \\ 2 & 3 \end{vmatrix} - 1 \cdot \begin{vmatrix} 2 & 1 \\ 0 & 3 \end{vmatrix} + 0 \cdot \begin{vmatrix} 2 & 3 \\ 0 & 2 \end{vmatrix} = \\ &= 9 - 2 - (6 - 0) + 0 = 1 \end{aligned}$$

+ - +
- + -
+ - +

1.2.73

$$\begin{aligned} \begin{vmatrix} -2 & 3 & 5 \\ 4 & 1 & -2 \\ 1 & -3 & 2 \end{vmatrix} &= 1 \cdot \begin{vmatrix} 3 & 5 \\ 1 & -2 \end{vmatrix} + 3 \cdot \begin{vmatrix} -2 & 5 \\ 4 & -2 \end{vmatrix} + 2 \cdot \begin{vmatrix} -2 & 3 \\ 4 & 1 \end{vmatrix} = \\ &= -6 - 5 + 3 \cdot (4 - 20) + 2 \cdot (-2 - 12) = \end{aligned}$$

+ - +
- + -
+ - +

$$= -11 - 48 - 28 = -87$$

1.2.74

$$\begin{aligned} + - + \rightarrow \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} &= a \cdot \begin{vmatrix} c & a \\ a & b \end{vmatrix} - b \cdot \begin{vmatrix} b & a \\ c & b \end{vmatrix} + c \cdot \begin{vmatrix} b & c \\ c & a \end{vmatrix} = \\ &= a(bc - a^2) - b(b^2 - ac) + c(ab - c^2) = \\ &= abc - a^3 - b^3 + abc + abc - c^3 = \\ &= 3abc - a^3 - b^3 - c^3 \end{aligned}$$

1.2.75

$$\begin{aligned} \begin{vmatrix} 2 & 0 & 0 \\ 0 & \beta & 0 \\ 0 & 0 & \gamma \end{vmatrix} &= 2\beta\gamma + 0 + 0 - 0 \cdot \beta \cdot 0 - 2 \cdot 0 \cdot 0 - \gamma \cdot 0 \cdot 0 = \\ &= 2\beta\gamma + 0 + 0 - 0 - 0 - 0 = 2\beta\gamma \end{aligned}$$

1.2.76

$$\begin{aligned} \begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix} &= 0 \cdot 0 \cdot 0 + 1 \cdot 1 \cdot 1 + 1 \cdot 1 \cdot 1 - 1 \cdot 0 \cdot 1 - 0 \cdot 1 \cdot 1 - 0 \cdot 1 \cdot 1 = \\ &= 0 + 1 + 1 - 0 - 0 - 0 = 2 \end{aligned}$$

1.2.77

$$\begin{aligned} \begin{vmatrix} \cos \alpha & \cos \beta & 0 \\ \cos \alpha & 0 & \cos \gamma \\ 0 & \cos \beta & \cos \gamma \end{vmatrix} &= \cos \alpha \cdot 0 \cdot \cos \gamma + 0 \cdot \cos \alpha \cdot \cos \beta + \\ &+ 0 \cdot \cos \beta \cdot \cos \gamma - 0 \cdot 0 \cdot 0 - \cos \alpha \cdot \cos \beta \cdot \cos \gamma - \\ &- \cos \gamma \cdot \cos \alpha \cdot \cos \beta = -2 \cos \alpha \cos \beta \cos \gamma \end{aligned}$$

1.2.78

$$\begin{vmatrix} 0 & x & 0 \\ x & 1 & x \\ 0 & x & 0 \end{vmatrix} = 0 \cdot 1 \cdot 0 + 0 \cdot x \cdot x + 0 \cdot x \cdot x - 0 \cdot 1 \cdot 1 - 0 \cdot x \cdot x - 0 \cdot x \cdot x = 0 + 0 + 0 - 0 - 0 - 0 = 0$$

1.2.79

$$\begin{vmatrix} 2 & 3 & 5 \\ 0 & 1 & 0 \\ 6 & 7 & 8 \end{vmatrix} = 0 \cdot \begin{vmatrix} 3 & 5 \\ 7 & 8 \end{vmatrix} - 1 \cdot \begin{vmatrix} 2 & 5 \\ 6 & 7 \end{vmatrix} + 0 \cdot \begin{vmatrix} 2 & 3 \\ 6 & 7 \end{vmatrix} = 0 - 1 \cdot (16 - 30) - 0 = 14$$

1.2.80

$$\begin{vmatrix} 1 & 2 & 0 \\ 3 & 4 & 0 \\ 5 & 6 & 7 \end{vmatrix} = 0 \cdot \begin{vmatrix} 3 & 4 \\ 5 & 6 \end{vmatrix} - 0 \cdot \begin{vmatrix} 1 & 2 \\ 5 & 6 \end{vmatrix} + 7 \cdot \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = 0 - 0 + 7 \cdot (4 - 6) = -14$$

1.2.81

$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 0 \end{vmatrix} = 3 \cdot \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix} - 6 \cdot \begin{vmatrix} 1 & 2 \\ 7 & 8 \end{vmatrix} + 0 \cdot \begin{vmatrix} 1 & 2 \\ 4 & 5 \end{vmatrix} = 3 \cdot (32 - 35) - 6 \cdot (8 - 14) = -9 + 36 = 27$$

1.2.82

$$\begin{vmatrix} x & y & z \\ 0 & y & z \\ x & 0 & z \end{vmatrix} = x \cdot \begin{vmatrix} y & z \\ 0 & z \end{vmatrix} - 0 \cdot \begin{vmatrix} y & z \\ x & z \end{vmatrix} + x \cdot \begin{vmatrix} y & z \\ y & z \end{vmatrix} = x \cdot (yz - 0z) - 0 + x \cdot (yz - yz) = xyz - 0 + 0 = xyz$$

1.2.83.

$$\begin{vmatrix} \cos \alpha & \cos \beta & \cos \gamma \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{vmatrix} = \cos \alpha \cdot \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} - \cos \beta \cdot \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} + \cos \gamma \cdot \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} = \cos \alpha \cdot (1 - 0) - \cos \beta \cdot (1 - 1) + \cos \gamma \cdot (0 - 1) = \cos \alpha - \cos \beta - \cos \gamma$$

1.2.84.

$$\begin{vmatrix} -3 & 2 & 1 \\ x-1 & 0 & 7 \\ 2 & -1 & 3 \end{vmatrix} = 0$$

$$-3 \cdot 0 \cdot 3 + 2 \cdot 2 \cdot 7 + (x-1) \cdot (-1) - 1 \cdot 0 \cdot 2 - (-3) \cdot (-1) \cdot 7 - 3 \cdot 2 \cdot (x-1) = 0$$

$$28 - x + 1 - 21 - 6x + 6 = 0$$

$$-7x = -14$$

$$x = 2$$

1.2.85

$$\begin{vmatrix} 2 & 0 & -1 \\ 1 & x+5 & 2-x \\ 3 & -1 & 2 \end{vmatrix} \leq 4$$

$$2 \cdot (x+5) \cdot 2 + (-1) \cdot (-1) + 3 \cdot 2 \cdot 0 - (-1) \cdot (x+5) \cdot 3 - (2-x) \cdot (-1) \cdot 2 - 2 \cdot 0 \cdot 1 \leq 4$$

$$4x + 20 + 1 + 3x + 15 + 4 - 2x \leq 4$$

$$5x + 36 \leq 0$$

$$x \leq -7,2 \Rightarrow x \in (-\infty; -7,2]$$

1.2.86

$$\begin{vmatrix} x+2 & 4 & -1 \\ -2 & 2 & x-1 \\ 1 & 3 & 0 \end{vmatrix} = 0$$

$$(x+2) \cdot 2 \cdot 0 + (-1) \cdot (-2) \cdot 3 + 1 \cdot (-1) \cdot 4 - (-1) \cdot 2 \cdot 1 - 0 \cdot (-2) \cdot 4 - (x+2) \cdot (x-1) \cdot 3 = 0$$

$$6 + 4x - 4 + 2 - 3(x^2 - x + 2x - 2) = 0$$

$$4x + 4 - 3x^2 - 3x + 6 = 0$$

$$-3x^2 + x + 10 = 0 \quad | \cdot (-1)$$

$$3x^2 - x - 10 = 0$$

$$D = 1 + 120 = 121$$

$$x_{1,2} = \frac{1 \pm 11}{6} = \begin{cases} 2 \\ -\frac{5}{3} \end{cases}$$

1.2.87

$$\begin{vmatrix} -3 & x-1 & 1 \\ x+2 & 2 & 3 \\ 0 & 1 & x \end{vmatrix} = 6$$

$$-3 \cdot 2 \cdot x + 1 \cdot (x+2) \cdot 1 + 0 \cdot (x-1) \cdot 3 - 1 \cdot 2 \cdot 0 - (-3) \cdot 3 \cdot 1 - x \cdot (x+2) \cdot (x-1) = 6$$

$$-6x + x + 2 + 9 - x(x^2 - x + 2x - 2) = 6$$

$$-5x + 11 - x^3 - x^2 + 2x - 6 = 0$$

$$-x^3 - x^2 - 3x + 5 = 0$$

$$-x^3 + x^2 - 2x^2 + 2x - 5x + 5 = 0$$

$$-x^2(x-1) - 2x(x-1) - 5(x-1) = 0$$

$$-(x-1)(x^2 + 2x + 5) = 0 \quad | \cdot (-1)$$

$$(x-1)(x^2 + 2x + 5) = 0$$

$$\begin{aligned} x-1 &= 0 & x^2 + 2x + 5 &= 0 \\ x &= 1 & D &= 4 - 20 = -16 < 0 \Rightarrow x \notin \mathbb{R} \end{aligned}$$

1.2.88

$$\begin{vmatrix} 3 & 2 & -1 \\ x+2 & 0 & 1 \\ -2 & 3-x & 1 \end{vmatrix} < 0$$

$$3 \cdot 0 \cdot 1 + (x+2) \cdot (3-x) \cdot (-1) + (-2) \cdot 2 \cdot 1 - (-1) \cdot 0 \cdot (-2) - 1 \cdot (x+2) \cdot 2 - 3 \cdot (3-x) \cdot 1 < 0$$

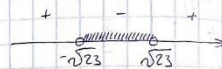
$$(x+2)(3-x) - 4 - 2x - 4 - 9 + 3x < 0$$

$$x^2 - 3x + 2x - 6 - 4 - 2x - 4 - 9 + 3x < 0$$

$$x^2 - 23 < 0$$

$$(x + \sqrt{23})(x - \sqrt{23}) < 0$$

$$x \in (-\sqrt{23}; \sqrt{23})$$

1.2.89 1) $(a-b)$, 2) $(b-c)$, 3) $(c-a)$

$$\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$$

$$1) \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} \xrightarrow{I-II} \begin{vmatrix} 0 & a-b & a^2-b^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (a-b) \begin{vmatrix} 0 & 1 & a-b \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$$

$$2) \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} \xrightarrow{II-III} \begin{vmatrix} 1 & a & a^2 \\ 0 & b-c & b^2-c^2 \\ 1 & c & c^2 \end{vmatrix} = (b-c) \begin{vmatrix} 1 & a & a^2 \\ 0 & 1 & b-c \\ 1 & c & c^2 \end{vmatrix}$$

$$3) \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} \xrightarrow{\text{III}-\text{I}} \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 0 & c-a & c^2-a^2 \end{vmatrix} = (c-a) \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 0 & 1 & c-a \end{vmatrix}$$

1.2.90 1)/(a-b), 2)/(b-c), 3)/(c-a)

$$\begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix}$$

$$1) \begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix} \xrightarrow{\text{I}-\text{II}} \begin{vmatrix} 0 & a-b & -c(a-b) \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix} = (a-b) \begin{vmatrix} 0 & 1 & -c \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix}$$

$$2) \begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix} \xrightarrow{\text{II}-\text{III}} \begin{vmatrix} 1 & a & bc \\ 0 & b-c & -a(b-c) \\ 1 & c & ab \end{vmatrix} = (b-c) \begin{vmatrix} 1 & a & bc \\ 0 & 1 & -a \\ 1 & c & ab \end{vmatrix}$$

$$3) \begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix} \xrightarrow{\text{III}-\text{I}} \begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 0 & c-a & -b(c-a) \end{vmatrix} = (c-a) \begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 0 & 1 & -b \end{vmatrix}$$

1.2.91 1)/(a-b), 2)/(b-c), 3)/(c-a)

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix}$$

$$1) \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} \xrightarrow{\text{I}-\text{II}} \begin{vmatrix} 0 & 1 & 1 \\ a-b & b & c \\ a^3-b^3 & b^3 & c^3 \end{vmatrix} = (a-b) \begin{vmatrix} 0 & 1 & 1 \\ 1 & b & c \\ a^2+ab+b^2 & b^3 & c^3 \end{vmatrix}$$

$$2) \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} \xrightarrow{\text{II}-\text{III}} \begin{vmatrix} 1 & 0 & 1 \\ a & b-c & c \\ a^3 & b^3-c^3 & c^3 \end{vmatrix} = (b-c) \begin{vmatrix} 1 & 0 & 1 \\ a & 1 & c \\ a^3 & b^2+bc+c^2 & c^3 \end{vmatrix}$$

$$3) \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} \xrightarrow{\text{III}-\text{I}} \begin{vmatrix} 1 & 1 & 0 \\ a & b & c-a \\ a^3 & b^3 & c^2-a^2 \end{vmatrix} = (c-a) \begin{vmatrix} 1 & 1 & 0 \\ a & b & c \\ a^3 & b^3 & c^2+ac+a^2 \end{vmatrix}$$

1.2.92

$$\begin{vmatrix} \sin x & \cos x & \sin(x+y) \\ \sin \beta & \cos \beta & \sin(\beta+y) \\ \sin \varphi & \cos \varphi & \sin(\varphi+y) \end{vmatrix} = \begin{vmatrix} \sin x & \cos x & \sin x \cos y + \cos x \sin y \\ \sin \beta & \cos \beta & \sin \beta \cos y + \cos \beta \sin y \\ \sin \varphi & \cos \varphi & \sin \varphi \cos y + \cos \varphi \sin y \end{vmatrix} =$$

$$= \begin{vmatrix} \sin x & \cos x & \sin x \cos y \\ \sin \beta & \cos \beta & \sin \beta \cos y \\ \sin \varphi & \cos \varphi & \sin \varphi \cos y \end{vmatrix} + \begin{vmatrix} \sin x & \cos x & \cos x \sin y \\ \sin \beta & \cos \beta & \cos \beta \sin y \\ \sin \varphi & \cos \varphi & \cos \varphi \sin y \end{vmatrix} =$$

$$= \cos y \begin{vmatrix} \sin x & \cos x \\ \sin \beta & \cos \beta \\ \sin \varphi & \cos \varphi \end{vmatrix} + \sin y \begin{vmatrix} \sin x & \cos x \\ \sin \beta & \cos \beta \\ \sin \varphi & \cos \varphi \end{vmatrix} =$$

$$= \cos y \cdot 0 + \sin y \cdot 0 = 0 + 0 = 0$$

1.2.93

$$\begin{vmatrix} a & a^2+1 & (a+1)^2 \\ b & b^2+1 & (b+1)^2 \\ c & c^2+1 & (c+1)^2 \end{vmatrix} = \begin{vmatrix} a & a^2+1 & a^2+2a+1 \\ b & b^2+1 & b^2+2b+1 \\ c & c^2+1 & c^2+2c+1 \end{vmatrix} =$$

$$= \begin{vmatrix} a & a^2+1 & a^2+1 \\ b & b^2+1 & b^2+1 \\ c & c^2+1 & c^2+1 \end{vmatrix} + \begin{vmatrix} a & a^2+1 & 2a \\ b & b^2+1 & 2b \\ c & c^2+1 & 2c \end{vmatrix} =$$

$$= 0 + 2 \begin{vmatrix} a & a^2+1 \\ b & b^2+1 \\ c & c^2+1 \end{vmatrix} = 0 + 2 \cdot 0 = 0 + 0 = 0$$

1.2.94

$$\begin{array}{c|ccc|ccc} x & a & b & 0 & c & + & - & + & - & + \\ 0 & y & 0 & 0 & d & - & + & - & + & - \\ 0 & e & z & 0 & 8 & + & - & + & - & + \\ \rightarrow 0 & 0 & 0 & 0 & v & - & + & - & + & - \\ g & h & k & u & l & + & - & + & - & + \end{array} = -v \cdot \begin{array}{c|ccc} x & a & b & 0 \\ 0 & y & 0 & 0 \\ 0 & e & z & 0 \end{array} = -v \cdot u \cdot x \cdot \begin{array}{c|ccc} y & 0 & 0 & 0 \\ 0 & e & z & 0 \\ g & h & k & u \end{array} = -v \cdot u \cdot x \cdot (yz - 0e) =$$

$$= -v \cdot u \cdot x \cdot yz$$

$$= -v \cdot u \cdot x \cdot yz$$

1.2.95

$$\begin{array}{c|ccc|ccc} 0 & 5 & 2 & 0 & + & - & + & - \\ 8 & 3 & 5 & 4 & - & + & - & + \\ 7 & 2 & 4 & 1 & + & - & + & - \\ \rightarrow 0 & 4 & 1 & 0 & + & - & + & - \end{array} = 4 \cdot \begin{array}{c|ccc} 0 & 2 & 0 & - \\ 8 & 5 & 4 & - \\ 7 & 4 & 1 & - \end{array} = 4 \cdot \begin{array}{c|ccc} 0 & 5 & 0 & + \\ - & 8 & 3 & 4 \\ + & - & - & - \\ + & - & + & + \end{array} =$$

$$= 4 \cdot (2 \cdot \begin{vmatrix} 8 & 4 \\ 7 & 1 \end{vmatrix} + 5 \cdot \begin{vmatrix} 8 & 4 \\ 7 & 1 \end{vmatrix} - 8 \cdot (8 \cdot 28) + 5 \cdot (8 \cdot 28)) =$$

$$= (-8 + 5) \cdot (8 \cdot 28) = -3 \cdot (-20) = 60$$

1.2.96

$$\begin{array}{c|ccc|ccc} 5 & a & 2 & -1 & + & - & + & - \\ 4 & b & 4 & -3 & - & + & - & + \\ 2 & c & 3 & -2 & + & - & + & - \\ 4 & d & 5 & -4 & - & + & - & + \end{array} = 5 \cdot \begin{array}{c|ccc} b & 4 & -3 \\ c & 3 & -2 \\ d & 5 & -4 \end{array} - a \cdot \begin{array}{c|ccc} 4 & 4 & -3 \\ 2 & 3 & -2 \\ 4 & 5 & -4 \end{array} + 2 \cdot \begin{array}{c|ccc} 4 & b & -3 \\ 2 & c & -2 \\ 4 & d & -4 \end{array} -$$

$$-(-1) \cdot \begin{vmatrix} 4 & b & 4 \\ 2 & c & 3 \\ 4 & d & 5 \end{vmatrix} =$$

$$= 5(b \cdot 3 \cdot (-4) + 4 \cdot (-2) \cdot d + 5c \cdot (-3) - 3 \cdot (-3) \cdot d - 4 \cdot (-4) \cdot c - (-2) \cdot 5 \cdot b) -$$

$$- a(4 \cdot 3 \cdot (-4) + 4 \cdot (-2) \cdot 4 + 2 \cdot 5 \cdot (-3) - 4 \cdot 3 \cdot (-3) - 4 \cdot 2 \cdot (-4) \cdot 5 \cdot (-2) \cdot 4) +$$

$$+ 2(4 \cdot (-4) \cdot c + 2 \cdot (-3) \cdot d + 4 \cdot (-2) \cdot b - 4 \cdot (-3) \cdot c - 2 \cdot (-4) \cdot b - 4 \cdot (-2) \cdot d) +$$

$$+ 4 \cdot 5 \cdot c + 2 \cdot 4 \cdot d + 4 \cdot 3 \cdot b - 4 \cdot 4 \cdot c - 2 \cdot 5 \cdot b - 3 \cdot 4 \cdot d =$$

$$= 5(-12b - 8d - 15c + 9d + 16c + 10b) -$$

$$- a(-48 - 32 - 30 + 36 + 32 - 40) +$$

$$+ 2(-16c - 8b - 6d + 12c + 8b + 8d) +$$

$$+ 20c + 8d + 12b - 16c - 10b - 12d =$$

$$= 5(-2b + d + c) - a(-2) + 2(-4c + 2d) + 4c - 4d + 2b =$$

$$= -10b + 5d + 5c + 2a - 8c + 4d + 4c - 4d + 2b =$$

$$= 2a + c - 8b + 5d = 2a - 8b + c + 5d$$

1.2.97

$$\begin{array}{c|ccc|ccc} 3 & 2 & 2 & 2 & + & - & + & - \\ 9 & -8 & 5 & 10 & - & + & - & + \\ 5 & -8 & 5 & 8 & + & - & + & - \\ \rightarrow 6 & -5 & 4 & 7 & - & + & - & + \end{array} = 3 \cdot \begin{array}{c|ccc} -8 & 5 & 10 \\ -8 & 5 & 8 \\ -5 & 4 & 7 \end{array} - 2 \cdot \begin{array}{c|ccc} 9 & 5 & 10 \\ 5 & 5 & 8 \\ 6 & 4 & 7 \end{array} + 2 \cdot \begin{array}{c|ccc} 9 & -8 & 10 \\ 5 & -8 & 8 \\ 6 & -5 & 7 \end{array} -$$

$$-2 \cdot \begin{vmatrix} 9 & -8 & 5 \\ 5 & -8 & 5 \\ 6 & -5 & 4 \end{vmatrix} =$$

$$= 3((-8) \cdot 5 \cdot 7 + 5 \cdot 8 \cdot (-5) + (-8) \cdot 4 \cdot 10 - 10 \cdot 5 \cdot (-5) - 5 \cdot (-8) \cdot 7 - 4 \cdot 8 \cdot (-8)) -$$

$$- 2(9 \cdot 5 \cdot 7 + 5 \cdot 8 \cdot 6 + 5 \cdot 4 \cdot 10 - 6 \cdot 5 \cdot 10 - 5 \cdot 5 \cdot 7 - 4 \cdot 8 \cdot 9) +$$

$$+ 2(9 \cdot (-8) \cdot 7 + (-8) \cdot 8 \cdot 6 + 5 \cdot (-5) \cdot 10 - 10 \cdot (-8) \cdot 6 - 5 \cdot (-8) \cdot 7 - 8 \cdot (-5) \cdot 9) -$$

$$\begin{aligned}
 & -2(9 \cdot 8) \cdot 4 + 5 \cdot (-8) \cdot 6 + 5 \cdot (-5) \cdot 5 - 5 \cdot (-8) \cdot 6 - 5 \cdot (-8) \cdot 4 - 5 \cdot (-5) \cdot 9) = \\
 & = 5(-280 + (-200) + (-320) - (-250) - (-280) - (-250)) - \\
 & \quad -2(515 + 240 + 200 - 300 - 175 - 268) + \\
 & \quad + 2(-504 + (-384) + (-250) - (-480) - (-280) - (-360)) - \\
 & \quad -2(-288 + (-240) + (-125) - (-240) - (-160) - (-225)) = \\
 & = 5(-14) - 2(-8) + 2(-18) - 2(-28) = \\
 & = -42 + 16 - 36 + 56 = -6
 \end{aligned}$$

1.2.98

$$\rightarrow \begin{vmatrix} 7 & 3 & 2 & 6 \\ 8 & -9 & 4 & 9 \\ 7 & -2 & 7 & 3 \\ 5 & -3 & 3 & 4 \end{vmatrix} \begin{matrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{matrix} =$$

$$\begin{aligned}
 & = 7 \begin{vmatrix} 9 & 4 & 9 \\ -2 & 7 & 3 \\ -3 & 3 & 4 \end{vmatrix} - 3 \begin{vmatrix} 8 & 4 & 9 \\ 7 & 7 & 3 \\ 5 & 3 & 4 \end{vmatrix} + 2 \begin{vmatrix} 8 & -9 & 9 \\ 7 & -2 & 3 \\ 5 & -3 & 4 \end{vmatrix} - 6 \begin{vmatrix} 8 & -9 & 4 \\ 7 & -2 & 7 \\ 5 & -3 & 3 \end{vmatrix} = \\
 & = -280 + 78 + 52 + 300 = 150
 \end{aligned}$$

$$\begin{aligned}
 1) & 7((-9) \cdot 7 \cdot 4 + 4 \cdot 3 \cdot (-3) + 5 \cdot (-2) \cdot 9 - 9 \cdot (-3) \cdot 7 - 4 \cdot (-2) \cdot 4 - 3 \cdot (-9) \cdot 3) = \\
 & = 7(-252 - 36 - 54 + 189 + 32 + 81) = 7 \cdot (-40) = -280
 \end{aligned}$$

$$\begin{aligned}
 2) & 3(8 \cdot 7 \cdot 4 + 4 \cdot 3 \cdot 5 + 7 \cdot 3 \cdot 9 - 9 \cdot 7 \cdot 5 - 7 \cdot 4 \cdot 4 - 3 \cdot 3 \cdot 8) = \\
 & = 3(224 + 60 + 189 - 315 - 112 - 72) = 3 \cdot (-26) = -78
 \end{aligned}$$

$$\begin{aligned}
 3) & 2(8 \cdot (-2) \cdot 4 + 3 \cdot (-9) \cdot 5 + 7 \cdot (-3) \cdot 9 - 9 \cdot (-2) \cdot 5 - 7 \cdot (-9) \cdot 4 - 3 \cdot (-3) \cdot 8) = \\
 & = 2(-64 - 135 - 189 + 90 + 252 + 72) = 2 \cdot 26 = 52
 \end{aligned}$$

$$\begin{aligned}
 4) & 6(8 \cdot (-2) \cdot 3 + 7 \cdot (-9) \cdot 5 + 7 \cdot (-3) \cdot 4 - 4 \cdot (-2) \cdot 5 - 7 \cdot (-9) \cdot 3 - 7 \cdot (-3) \cdot 8) = \\
 & = 6(-48 - 315 - 84 + 40 + 189 + 168) = 6 \cdot (-50) = -300
 \end{aligned}$$

1.2.99

$$\rightarrow \begin{vmatrix} 3 & 6 & 5 & 6 & 4 \\ 5 & 9 & 7 & 8 & 6 \\ 6 & 12 & 13 & 9 & 7 \\ 4 & 6 & 6 & 5 & 4 \\ 2 & 5 & 4 & 5 & 3 \end{vmatrix} \begin{matrix} + & - & + & - & + \\ - & + & - & + & - \\ + & - & + & - & + \\ - & + & - & + & - \\ + & - & + & - & + \end{matrix} =$$

$$\begin{aligned}
 & = 3 \begin{vmatrix} 9 & 7 & 8 & 6 \\ 12 & 13 & 9 & 7 \\ 6 & 6 & 5 & 4 \\ 5 & 4 & 5 & 3 \end{vmatrix} - 6 \begin{vmatrix} 5 & 7 & 8 & 6 \\ 6 & 13 & 9 & 7 \\ 4 & 6 & 5 & 4 \\ 2 & 4 & 5 & 3 \end{vmatrix} + 5 \begin{vmatrix} 5 & 9 & 8 & 6 \\ 6 & 12 & 9 & 7 \\ 4 & 6 & 5 & 4 \\ 2 & 5 & 5 & 3 \end{vmatrix} - 6 \begin{vmatrix} 5 & 9 & 8 & 6 \\ 6 & 12 & 9 & 7 \\ 4 & 6 & 5 & 4 \\ 2 & 5 & 5 & 3 \end{vmatrix} =
 \end{aligned}$$

$$\begin{aligned}
 & + 6 \begin{vmatrix} 5 & 9 & 7 & 8 \\ 6 & 12 & 13 & 9 \\ 4 & 6 & 6 & 5 \\ 2 & 5 & 4 & 3 \end{vmatrix} + 4 \begin{vmatrix} 5 & 9 & 7 & 8 \\ 6 & 12 & 13 & 9 \\ 4 & 6 & 6 & 5 \\ 2 & 5 & 4 & 3 \end{vmatrix} = \times
 \end{aligned}$$

$$\begin{aligned}
 1) & \rightarrow \begin{vmatrix} 9 & 7 & 8 & 6 \\ 12 & 13 & 9 & 7 \\ 6 & 6 & 5 & 4 \\ 5 & 4 & 5 & 3 \end{vmatrix} \begin{matrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{matrix} =
 \end{aligned}$$

$$\begin{aligned}
 & = 9 \begin{vmatrix} 13 & 9 & 7 \\ 6 & 5 & 4 \\ 4 & 5 & 3 \end{vmatrix} - 7 \begin{vmatrix} 12 & 9 & 7 \\ 6 & 5 & 4 \\ 5 & 5 & 3 \end{vmatrix} + 8 \begin{vmatrix} 12 & 13 & 7 \\ 6 & 6 & 4 \\ 5 & 4 & 3 \end{vmatrix} - 6 \begin{vmatrix} 12 & 13 & 9 \\ 6 & 6 & 5 \\ 5 & 4 & 5 \end{vmatrix} =
 \end{aligned}$$

$$= 9(13 \cdot 5 \cdot 3 + 9 \cdot 4 \cdot 4 + 6 \cdot 5 \cdot 7 - 4 \cdot 5 \cdot 7 - 6 \cdot 9 \cdot 3 - 5 \cdot 4 \cdot 13) -$$

$$\begin{aligned}
 &= 5(12 \cdot 6 \cdot 5 + 6 \cdot 4 \cdot 9 + 13 \cdot 5 \cdot 5 - 9 \cdot 6 \cdot 5 - 4 \cdot 5 \cdot 12 - 6 \cdot 13 \cdot 5) - \\
 &\quad - 9(6 \cdot 6 \cdot 5 + 4 \cdot 4 \cdot 9 + 13 \cdot 5 \cdot 2 - 9 \cdot 6 \cdot 2 - 4 \cdot 5 \cdot 6 - 4 \cdot 13 \cdot 5) + \\
 &\quad + 7(6 \cdot 6 \cdot 5 + 4 \cdot 5 \cdot 9 + 12 \cdot 5 \cdot 2 - 9 \cdot 6 \cdot 2 - 5 \cdot 5 \cdot 6 - 4 \cdot 12 \cdot 5) - \\
 &\quad - 8(6 \cdot 6 \cdot 4 + 4 \cdot 5 \cdot 13 + 12 \cdot 6 \cdot 2 - 13 \cdot 6 \cdot 2 - 5 \cdot 6 \cdot 6 - 4 \cdot 12 \cdot 4) = \\
 &= 5 \cdot 1 - 9 \cdot (-34) + 7 \cdot (-18) - 8 \cdot 20 = 5 + 306 - 126 - 160 = 25.
 \end{aligned}$$

$$\begin{aligned}
 * &= 3 \cdot (-10) - 6 \cdot 15 + 5 \cdot 5 - 6 \cdot 0 + 4 \cdot 25 = \\
 &= -30 - 90 + 25 - 0 + 100 = \underline{5}
 \end{aligned}$$